

The inertia group is under-constrained

why the pure-symmetry route to “ $\mathcal{P}^{[d]G}$ unramified at η_C ” stalls, and where higher ramification has to enter

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Purpose. This note records, and explains the failure of, the first approach we tried for the unramifiedness theorem

$$(\mathbf{H1}) + (\mathbf{H2}) + (\mathbf{H3}) \implies \mathcal{P}^{[d]G} \rightarrow U^{(d)} \text{ unramified at } \eta_C.$$

The approach is attractive because it is purely formal: it stays inside the Galois group of the master diagram, encodes the whole theorem as a single containment of inertia groups, and never invokes local ramification theory. The outcome is sharp and negative.

The note contains *two* summaries of the same dead end, which may be read independently:

Part I (short form). The abstract reason it fails: (H1) and (H2) determine only two projections of the relevant inertia group, and by Goursat’s lemma a *diagonal* subgroup is left unconstrained – precisely the one that breaks the conclusion.

Part II (long form). The detailed working session of 2026-05-14: we computed the inertia group *outright* in the local model at the degenerate point $d \cdot p_\infty$, found the diagonal is genuinely populated, hit an apparent counterexample to the theorem as first stated, and isolated exactly what higher ramification must (and must not) be asked to do.

Throughout we assume the setup, notation and master diagram of the standalone proof note, recalling only what is needed. The common notation is fixed once, in [Section 1](#), and used by both parts.

1 Common set-up

Recall the four symmetric quotients $\mathcal{P}^{[d]G}$, $\mathcal{P}^{[d]H}$, $(\mathcal{P}_{G/H})^{(d)}$, $(\mathcal{P}_{G/H})^{[d]}$ and the master diagram with Cartesian upper square. All of them are quotients of the single connected G^d -cover $\mathcal{P}^{\times d} \rightarrow U^d$, so the whole picture is governed by one Galois extension. Writing

$$K := K(U^{(d)}) = K(C^{(d)}), \quad N := K(\mathcal{P}^{\times d}),$$

the extension N/K is Galois with group

$$\Gamma := G^d \rtimes S_d,$$

the S_d permuting the d factors of the diagonal G^d . With $K_G = \ker(G^d \xrightarrow{\Sigma} G)$ and $K_H = \ker(H^d \rightarrow H)$, the five fields are the fixed fields of subgroups of Γ :

$$\begin{aligned} L &:= K(\mathcal{P}^{[d]G}) & \leftrightarrow & \Xi := K_G \rtimes S_d, & \text{Gal}(L/K) &\cong G, \\ M &:= K((\mathcal{P}_{G/H})^{[d]}) & \leftrightarrow & (H^d + K_G) \rtimes S_d, & \text{Gal}(M/K) &\cong G/H, \\ K' &:= K((\mathcal{P}_{G/H})^{(d)}) = K(C'^{(d)}) & \leftrightarrow & H^d \rtimes S_d, \\ L_H &:= K(\mathcal{P}^{[d]H}) & \leftrightarrow & K_H \rtimes S_d. \end{aligned}$$

The diagonal map L/K has group G via summation $\Sigma : \Gamma \rightarrow G$, $((g_i), \sigma) \mapsto \sum_i g_i$; the conclusion we want is that a certain inertia group lands in $\ker \Sigma = K_G \rtimes S_d$. The four fields M, K', L, L_H sit in a *parallelogram* over M : one checks $M \subseteq K'$, $L \cap K' = M$ and $L \cdot K' = L_H$, so L/M and K'/M are linearly disjoint with compositum L_H and

$$\text{Gal}(L_H/M) = \underbrace{\text{Gal}(L/M)}_A \oplus \underbrace{\text{Gal}(K'/M)}_B. \quad (1)$$

Fix a discrete valuation $\widetilde{W}$ on N extending V_C , chosen so that $\widetilde{W}|_{K'} = V_{C'}$ (possible because Γ acts transitively on the valuations above V_C), and let

$$J := I(\widetilde{W}/V_C) \subseteq \Gamma$$

be its inertia group. The conclusion of the theorem is exactly the single containment

$$J \subseteq \Xi = K_G \rtimes S_d, \quad \text{equivalently} \quad \sum_i g_i = 0 \text{ in } G \quad \forall ((g_i), \sigma) \in J, \quad (2)$$

since the image of J in G under Σ is the inertia of $\widetilde{W}|_L$ over V_C .

Part I

Short form: the Goursat gap

2 Three projections of one inertia group

We are forced up to N by the *hypotheses*, not the conclusion: (H1) lives in $\text{Gal}(M/K) = G/H$, (H2) in $\text{Gal}(L_H/K') = H$, the goal in $\text{Gal}(L/K) = G$ – three different quotients of Γ with no common Galois field below N . So we read three projections off the single inertia group J :

projection of J	equals inertia of	status
$J \xrightarrow{\Sigma} G$	$\widetilde{W} _L$ over V_C	the goal
$J \xrightarrow{\Sigma \bmod H} G/H$	$\widetilde{W} _M$ over V_C	trivial by (H1)
$J \cap (H^d \rtimes S_d) \xrightarrow{\Sigma} H$	$\widetilde{W} _{L_H}$ over $V_{C'}$	trivial by (H2)

In symbols,

$$(H1) : \quad \sum_i g_i \in H \quad \forall ((g_i), \sigma) \in J; \quad (3)$$

$$(H2) : \quad J \cap (H^d \rtimes S_d) \subseteq K_H \rtimes S_d, \quad \text{i.e. } \sum_i h_i = 0 \text{ whenever } ((h_i), \sigma) \in J, h_i \in H. \quad (4)$$

By (3) every element of J has $(g_i) \in H^d + K_G$; writing $g_i = h_i + k_i$ with $(h_i) \in H^d$, $(k_i) \in K_G$ (so $\sum_i k_i = 0$), the goal (2) becomes

$$\sum_i h_i = 0 \text{ in } H \quad \forall ((g_i), \sigma) \in J. \quad (5)$$

3 Where the symmetry runs out

Hypothesis (4) delivers (5) *only* for elements that already lie in H^d (the case $(k_i) = 0$). For a general element with $(k_i) \neq 0$ neither hypothesis says anything: (4) is about elements *inside* $H^d \rtimes S_d$, and (3) only sees the G/H -component, which is vacuous here.

Concretely, the obstruction is a **diagonal inertia subgroup**. Nothing in (3) and (4) prevents

$$J \supseteq \{ (h, \kappa(h)) : h \in H_0 \} \subseteq H \oplus K_{G/H}, \quad (6)$$

glued along a homomorphism $\kappa : H_0 \rightarrow K_{G/H}$: it projects trivially to G/H (so (3) holds) and meets $H^d \rtimes S_d$ only in $\kappa(h) = 0$ (so (4) holds vacuously), yet has $\sum_i h_i \neq 0$, breaking (5).

The structural reason. The symmetry available to us – the two quotient/restriction maps coming from $H \trianglelefteq G$ – only resolves J along the two coordinate axes of the split sequence $0 \rightarrow K_H \rightarrow K_G \rightarrow K_{G/H} \rightarrow 0$. An inertia group need not respect that splitting; it can sit diagonally across it. Knowing two coordinate projections of a subgroup of a product never determines the subgroup – this is exactly the content of Goursat’s lemma – so there is simply *not enough symmetry* in the discrete inertia data alone to collapse the diagonal. The same point inside the parallelogram (1): an inertia subgroup $\tilde{I} \subseteq A \oplus B$ compatible with (H1),(H2) can be either the harmless B (then L/M unramified, theorem holds) or the fatal diagonal Δ (then L/M ramified, theorem fails), and the hypotheses cannot tell them apart.

Part II

Long form: the explicit local computation

The short form leaves open whether the diagonal of (6) is a real possibility or an artefact of weak bookkeeping. In the session of 2026-05-14 we settled this by computing J outright in the local model. The answer: the diagonal is genuinely there, the theorem as first stated has a counterexample, and higher ramification enters in a specific, limited way.

4 Computing J in the local model

Assume $\mathcal{P} \rightarrow U$ is totally ramified at p_∞ with inertia $G_0 \subseteq G$ (else replace G by G_0). Choose a uniformiser t on C and u_j on $\overline{\mathcal{P}}$ with $t_j = u_j^{|G_0|} + \dots$. The symmetric local ring is $k[[e_1, \dots, e_d]]$

in the elementary symmetric functions of the t_j , and the geometrically correct valuation V_C is the *weighted* blowup with

$$w(e_i) = i, \quad \tilde{w}(u_j) = 1.$$

(With the ordinary unweighted blowup V_C does not even extend to $V_{C'}$.) Decompose J along $K \subseteq K(C^d) \subseteq N$.

The S_d -part is trivial. The residue field of V_C is that of the weighted projective space $\mathbb{P}(1, 2, \dots, d)$, and S_d acts faithfully on the residue field of the lift to $K(C^d)$ (it permutes the $\mathbb{P}^{d-1}$ -coordinates). So no nontrivial permutation lies in inertia: $J \rightarrow S_d$ is trivial, i.e. $J \subseteq G^d$.

The G^d -part, tame case: $J = G_{\text{diag}}$. In the tame case g_j acts by $u_j \mapsto \zeta^{g_j} u_j$ (ζ a primitive $|G_0|$ -th root of unity). On the residue field $k(\mathbb{P}_u^{d-1})$ the tuple $(g_1, \dots, g_d)$ acts by $(u_1 : \dots : u_d) \mapsto (\zeta^{g_1} u_1 : \dots : \zeta^{g_d} u_d)$, trivial in PGL_d iff $g_1 = \dots = g_d$. Hence

$$\boxed{J = G_{\text{diag}} = \{((g, g, \dots, g), e) : g \in G_0\}} \quad (\text{tame, totally ramified}). \quad (7)$$

The G^d -part, wild case: $J \supseteq G^d$. If G acts wildly with conductor $m \geq 1$ then $g_j(u_j) = u_j + a u_j^{m+1} + \dots$, and the correction has $\tilde{w}$ -value $\geq m+1 > 0$, hence *vanishes in the residue field*. So *every* tuple in G^d acts trivially on the residue of $\tilde{W}$: $J \supseteq G^d$, strictly larger than the diagonal.

Reading off the image in G . Under summation, $G_{\text{diag}} \rightarrow G$ is $(g, \dots, g) \mapsto d \cdot g$, so

$$\text{Im}(J \rightarrow G) = d \cdot G_0, \quad (8)$$

trivial iff $|G_0| \mid d$. The leading-order inertia at η_C is thus the *cardinality* condition $|G_0| \mid d$ – nothing about the depth of ramification.

5 The diagonal is genuinely there: K'/M is ramified

One might hope the diagonal is excluded by some automatic fact. It is not. The would-be saving hypothesis is

$$(H3') \quad K'/M \text{ unramified at } V_{C'}, \quad \text{i.e.} \quad J \cap \text{Gal}(N/M) \subseteq H^d \rtimes S_d,$$

which would force $(k_i) = 0$ in (5) and finish the proof. But the same weighted-blowup computation gives the inertia of K'/M explicitly. With $E = G/H$ acting on C' by $u \mapsto \zeta u$, an element $\vec{a} \in E_+^d = \ker(\Sigma : E^d \rightarrow E)$ acts by $\epsilon_i \mapsto e_i(\zeta^{a_1} u_1, \dots)$; the *diagonal* $\vec{a} = (a, \dots, a)$ scales $\epsilon_i \mapsto \zeta^{ia} \epsilon_i$ and so fixes every weight-zero coordinate $\xi_i = \epsilon_i / \epsilon_1^i$, while non-diagonal $\vec{a}$ act nontrivially. Hence

$$I_{V_{C'}}(K'/M) = E_{\text{diag}} \cap E_+^d \cong \ker(d : E \rightarrow E) \cong \mathbb{Z} / \gcd(d, |E|), \quad (9)$$

nontrivial as soon as $\gcd(d, |E|) > 1$ – e.g. *totally ramified* when $d = |E| = 2$. Two features make this fatal:

- **It depends only on E and d , not on G .** The map $q : (\mathcal{P}_{G/H})^{(d)} \rightarrow (\mathcal{P}_{G/H})^{[d]}$ sees only $\mathcal{P}_{G/H}$, so *no* hypothesis on the ambient G – cyclic, prime-power, or a ramification bound on $\mathcal{P}$ at p_∞ – can change (9).

- **The available lemma does not reach $V_{C'}$.** q is known unramified only at codimension-1 boundary valuations of $C'^{(d)}$, whereas $V_{C'}$ sits over the codimension- d point $d \cdot p'_\infty$, where all d points collide at the branch point and the diagonal E -action stabilises the configuration most strongly.

An apparent counterexample to the theorem-as-stated. Pushing through to L/K produced not just a stalled proof but a concrete obstruction. Take $\text{char} \neq 2$, $G = \mathbb{Z}/4$, $H = \mathbb{Z}/2$, $E = \mathbb{Z}/2$, $d = 2$, and $\mathcal{P} \rightarrow U$ tamely totally ramified of index 4 at p_∞ . Then (H3) holds (G cyclic 2^2), the ramification bound holds vacuously (tame: $G_1 = 0$, so $G_d = 0$ for all d), and one checks (H1), (H2) both hold. Yet $L = K(\sqrt[4]{e_2})$ with $v_C(e_2) = 2$, so $v_L(\sqrt[4]{e_2}) = \frac{1}{2}$ and L/K is *ramified* at V_C with inertia of order 2 – exactly the predicted $d \cdot G_0 = 2 \cdot \mathbb{Z}/4 = H$ of (8). The missing hypothesis is visibly $|G_0| \mid d$, which fails here ($4 \nmid 2$).

6 Why higher ramification is the right – and necessary – language

The inertia group is the bottom step $G^0 = G_0$ of the ramification filtration; it remembers *whether* a place ramifies but discards the *jumps*. The diagonal subgroup is invisible to (H1), (H2) precisely because those are statements about inertia, and inertia cannot separate the H -jump from the G/H -jump of a diagonally entangled extension. Two points pin down the role of higher ramification:

1. **Upper numbering is the natural bookkeeping.** The argument keeps passing to quotients $G \rightarrow G/H$, and the upper-numbering filtration is functorial for quotients via Herbrand, $(G/H)^v = G^v \cdot H/H$, whereas lower numbering is functorial for subgroups. So the bound should be read in *upper* numbering, $G^d = \{1\}$. This is exactly what makes the eventual proof (Herbrand-convexity) work: for cyclic $G = \mathbb{Z}/p^r$ the jumps form a single chain and

$$\varphi_G(m_r) = \varphi_{G/H}(m_{r-s}) + \frac{\varphi_H(m_r) - m_{r-s}}{p^{r-s}}$$

is a *convex combination*, so (H1),(H2) $\Rightarrow \varphi_G(m_r) < d$. Convexity bounds the mixed jump by the two pure jumps *no matter how the extension is entangled* – the very diagonal that defeated the inertia argument.

2. **But reparameterising alone is not the fix.** The leading-order obstruction (8) depends only on the *cardinality* $|G_0| = |G^0|$ relative to d , and upper vs. lower numbering are two parameterisations of the *same* filtration – they do not change G_0 , nor the diagonal $J = G_{\text{diag}}$ that summation sends to $d \cdot G_0$. The higher filtration *augments* the leading-order picture (it controls the next-order action on tangent and higher infinitesimal data at p'_∞ , hence the codimension at which the ramification of $\mathcal{P}$ becomes visible in $C^{(d)}$); it does not by itself collapse the leading-order diagonal. In the genuinely useful regime the bound $G^d = \{1\}$ is non-vacuous and is consumed by the Herbrand-convexity estimate, not by a leading-order count.

7 Summary

- **The route.** Encode the theorem as $J \subseteq K_G \rtimes S_d$ for the inertia group $J \subseteq \Gamma = G^d \rtimes S_d$, and read three projections of J off (H1), (H2) and the goal.

- **Why it stalls (short form).** (H1), (H2) fix only two projections; by Goursat a diagonal subgroup Δ is left unconstrained, and the hypotheses cannot tell the harmless inertia B from the fatal Δ .
- **The diagonal is real (long form).** The explicit weighted-blowup computation gives $J = G_{\text{diag}}$ (tame) or $J \supseteq G^d$ (wild), with $\text{Im}(J \rightarrow G) = d \cdot G_0$; and K'/M has inertia $\mathbb{Z}/\gcd(d, |E|)$, depending only on E, d . No hypothesis on G , and no ramification bound on $\mathcal{P}$, removes it; the theorem as first stated even has a tame counterexample, missing the hypothesis $|G_0| \mid d$.
- **The moral.** Discrete inertia data is too coarse. The missing rigidity lives in the *higher ramification filtration*: read the bound in upper numbering and feed it into the Herbrand-convexity lemma, which is how the successful proofs of the unramifiedness theorem close the gap. The inertia/symmetry argument is non-load-bearing – it cannot reach the conclusion on its own.